

Name: Abraham Abebe

Course: Cybersecurity

Date: July 29, 2026

Title: Social Engineering Toolkit (SET) and Reconnaissance Lab

Exercise 2: Advanced Phishing Simulation with Zphisher

Objective

The objective of this exercise was to examine how Zphisher can be used within a controlled laboratory environment to simulate phishing attacks using pre-built website templates. The exercise demonstrates how phishing frameworks automate the creation of realistic login pages and highlights the importance of user awareness and defensive security practices.

Environment

Item	Description
Host Operating System	Arch Linux (Main Boot OS)
Tool Used	Zphisher
Shell	Bash
Network	Local Laboratory Network
Purpose	Security Awareness and Defensive Training

Procedure

Step 1: Launch Zphisher

The Zphisher framework was launched from the terminal.

Command

```
bash zphisher.sh
```

```
[abrish@archlinux zphisher]$ ls
auth  Dockerfile  LICENSE  make-deb.sh  README.md  run-docker.sh  scripts  zphisher.sh
[abrish@archlinux zphisher]$ bash zphisher.sh_
```

Observation

The application started successfully and displayed the main menu containing numerous phishing templates for popular online services.

Observation

The application initialized the local server and generated a URL that could be accessed from the test system.

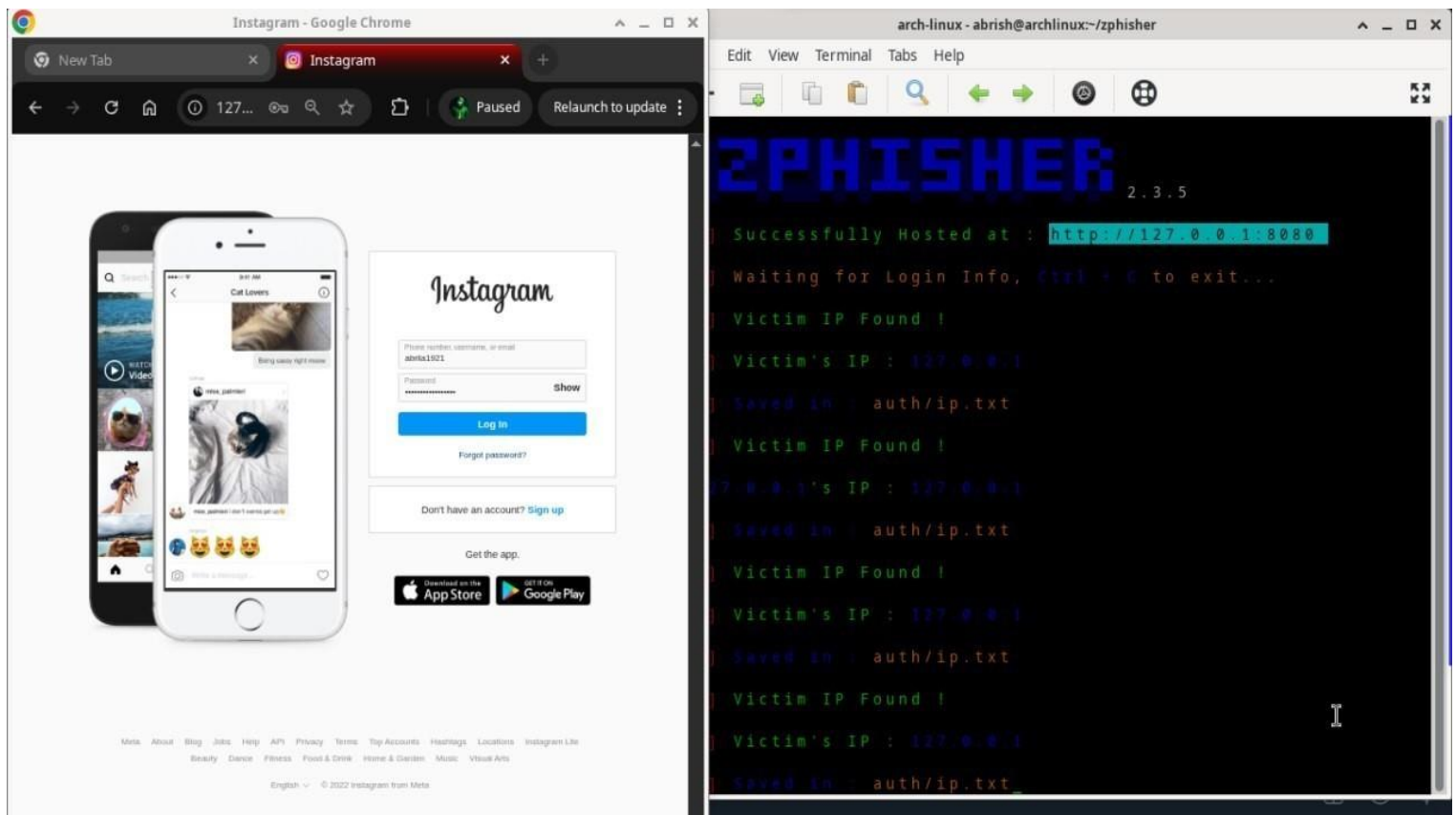


Figure 3: Local Hosting Configuration.

Step 4: Access the Simulated Login Page

The generated URL was opened in a web browser on the test machine.

Observation

The simulated login page closely resembled the legitimate website in terms of layout, colors, branding, and login interface.

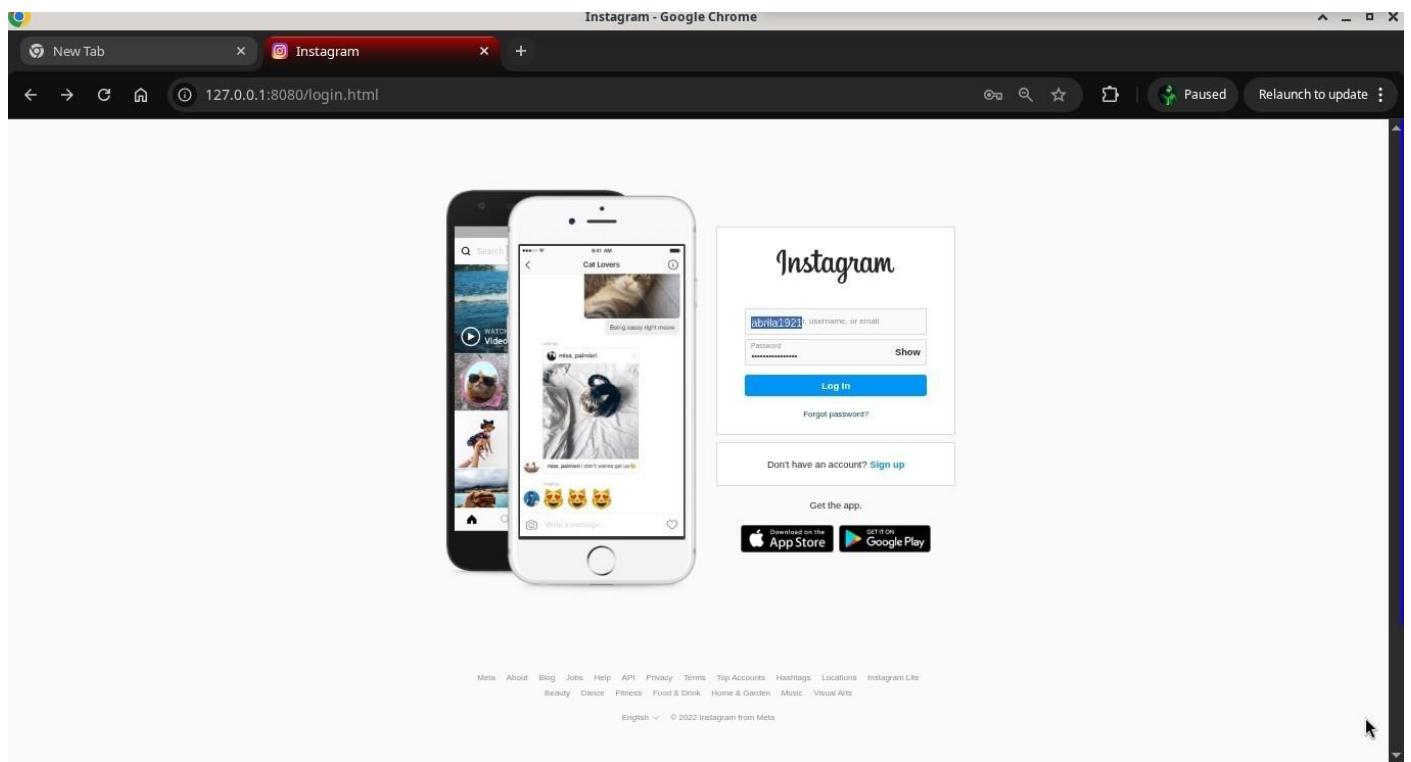


Figure 4: Simulated Login Page.

Step 5: Submit Test Credentials

Test credentials were entered into the simulated login page.

Observation

The login form accepted the submitted information and forwarded it to the Zphisher application for demonstration purposes.

```
ZPHISHER 2.3.5
[-] Successfully Hosted at : http://127.0.0.1:8080
[-] Waiting for Login Info, Ctrl + C to exit...
[-] Victim IP Found !
[-] Victim's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Victim IP Found !
[-] 127.0.0.1's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Victim IP Found !
[-] Victim's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Victim IP Found !
[-] Victim's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Login info Found !!
[-] Account : abril1921
[-] Password : abrahamabebe@1234
[-] Saved in : auth/usernames.dat
```

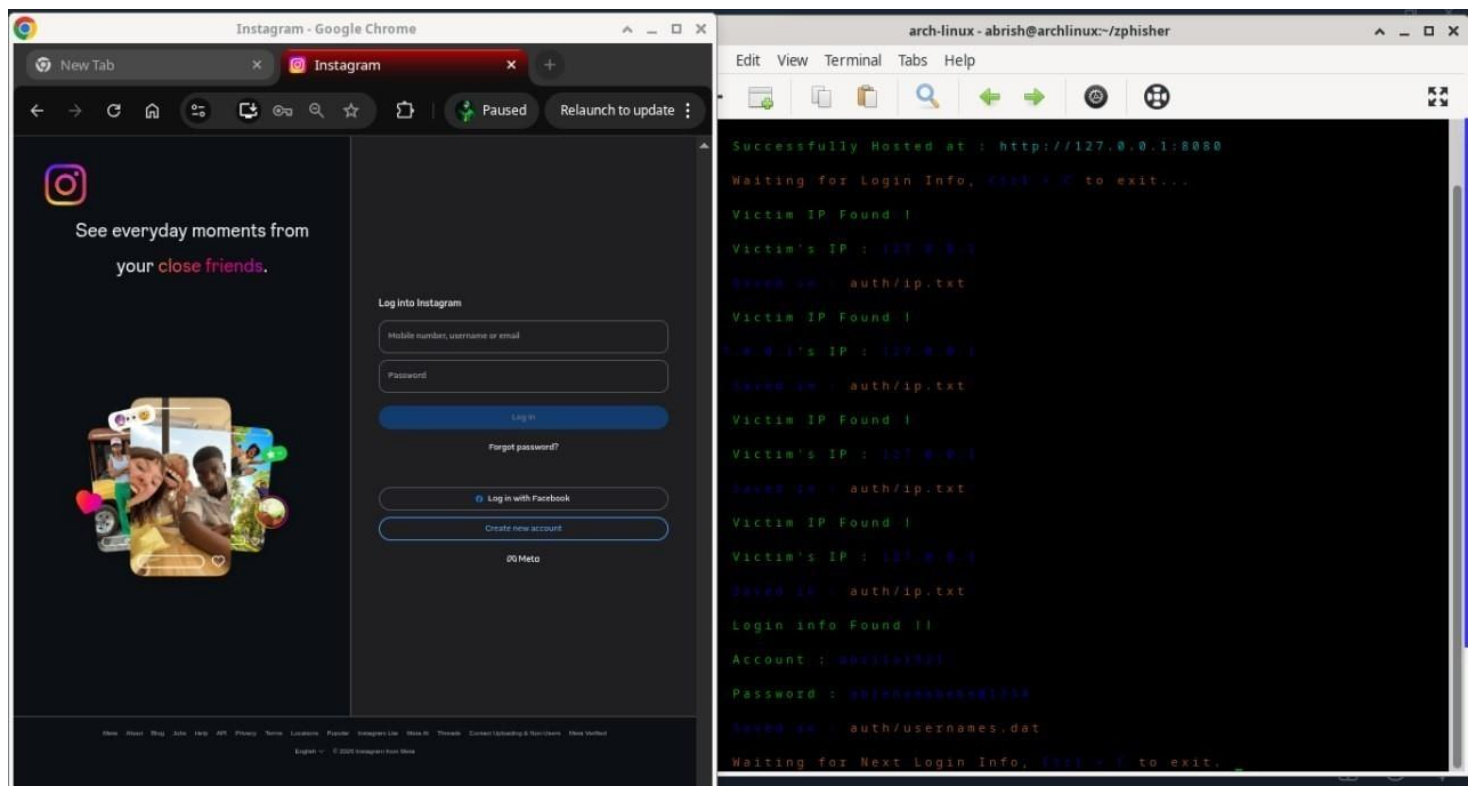


Figure 5: Submission of Test Credentials.

Step 6: Verify Captured Information

After the credentials were submitted, the terminal displayed the information captured during the simulation.

Observation

The application successfully recorded the submitted test credentials and associated connection information within the controlled laboratory environment.

```
[abrish@archlinux zphisher]$ ls
auth  Dockerfile  LICENSE  make-deb.sh  README.md  run-docker.sh  scripts  zphisher.sh
[abrish@archlinux zphisher]$ cat auth/usernames.dat

Instagram Username: abreham@1921 Pass: abreham@123456
Instagram Username: abril@1921 Pass: abrehamabebe1234
Instagram Username: abril@1921 Pass: abrehamabebe@1234
[abrish@archlinux zphisher]$ _
```

Figure 6: Captured Information Displayed by Zphisher.

Analysis Questions:

- How does Zphisher differ from SET in terms of templates?
- What makes these templates convincing?
- What defensive measures could detect these pages?

1. How does Zphisher differ from SET in terms of templates?

Zphisher provides a large collection of pre-built phishing templates for popular websites such as Facebook, Instagram, Google, and Netflix, allowing phishing pages to be deployed quickly with minimal configuration. In contrast, SET primarily uses website cloning, dynamically copying the target website rather than relying on pre-built templates. While Zphisher emphasizes speed and simplicity, SET offers greater flexibility by allowing users to clone virtually any website.

2. What makes these templates convincing?

Zphisher templates closely resemble legitimate websites by replicating their logos, colors, fonts, layouts, and login forms. Many templates also include realistic features such as responsive page design and multi-step login interfaces, making them appear authentic. This visual similarity can make it difficult for users to distinguish phishing pages from legitimate websites.

3. What defensive measures could detect these pages?

Organizations can detect and prevent phishing pages by implementing multiple security controls, including:

- Verifying website URLs and domain names before entering credentials.
- Using secure web gateways and DNS filtering to block malicious websites.
- Deploying browser anti-phishing protection and email filtering solutions.
- Enabling Multi-Factor Authentication (MFA), especially phishing-resistant methods such as FIDO2 or WebAuthn.
- Using password managers, which only autofill credentials on legitimate domains.
- Providing regular phishing awareness training so users can recognize suspicious websites and emails.

These defensive measures significantly reduce the likelihood of successful phishing attacks.

Analysis

This exercise demonstrated how Zphisher simplifies phishing website deployment using pre-built templates that closely resemble popular online services. Compared to manual website cloning, it enables faster and easier creation of convincing phishing pages. Unlike SET, which dynamically clones websites, Zphisher relies on pre-configured templates for rapid deployment. The exercise also emphasized that users should verify website addresses and avoid entering credentials on untrusted login pages.

Results

The phishing simulation was successfully completed within the isolated laboratory environment.

The following outcomes were observed:

- Zphisher launched successfully.

- The simulated login page was hosted locally.
- A phishing template was selected and configured.
- The generated URL was accessible from the test system.
- Test credentials were submitted successfully.
- The submitted information was captured and displayed by the application.
- The exercise demonstrated how phishing frameworks automate credential collection during security awareness simulations.

Defensive Countermeasures

To reduce phishing attacks, organizations should:

- Verify website URLs before entering credentials.
- Avoid relying only on HTTPS or the padlock icon.
- Enable Multi-Factor Authentication (MFA).
- Conduct regular phishing awareness training.
- Use secure email gateways with phishing filtering.
- Monitor login activity for suspicious behavior.
- Encourage users to report suspicious emails and websites.

Conclusion

This exercise demonstrated how Zphisher can simulate phishing attacks using realistic website templates in a controlled lab environment. It highlighted the importance of verifying website authenticity, using Multi-Factor Authentication, and improving user awareness. The exercise also showed that combining security training with technical controls is essential for reducing the success of phishing attacks.